



Technical Report – Infrared Receiver

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Photodiodes behavior (light sensor)

When illuminated by the infrared beam, the photodiode generates a photocurrent proportional to the light intensity. It is reverse-biased to widen the depletion region, reducing capacitance and improving both response speed and sensitivity. With the IR beam present, the resulting photocurrent produces a signal that is amplified and processed by the PLL. If the beam is blocked, the photocurrent disappears, no signal reaches the amplifier, the PLL loses lock, and the circuit triggers the alarm.

Transistor Amplifier Stage (signal Conditioning)

The two NPN transistors form a cascaded amplifier that boosts the small signal from the photodiode. A single stage would not provide enough gain, so two are used to ensure a strong and noise-resistant output for the PLL. Both transistors operate slightly into saturation, making the stage behave almost like a switch and producing sharp, stable waveform transitions. This results in a clean and consistent signal that can be easily detected by the next circuit block.

Phase-Locked Loop (NE/LM567) function

The PLL (Phase-Locked Loop) continuously compares the incoming signal frequency from the photodiode amplifier with its internally generated reference. When both frequencies match, the PLL achieves “lock,” meaning it has synchronized with the 4.5 kHz modulation transmitted by the IR source.

- Pin 8 LOW:
Indicates that the PLL is locked, confirming that the correct infrared signal is present and being received.
- Pin 8 HIGH:
Indicates that the PLL is not locked, meaning the 4.5 kHz tone is missing, typically because the beam is blocked. The system then triggers the alarm stage.

Thus, the PLL output serves as a digital indicator reflecting the presence or absence of the infrared signal.



Key formula for 567 Tone:

$$f_0 = \frac{1}{1,1 \times R_1 \times C_1}$$

From schematic:

$$R_1 = 10k\Omega$$

$$C_1 = 22 \times 10^{-9}F$$

Calculate 567 Tone frequency:

$$f_0 = \frac{1}{1,1 R_1 C_1}$$

$$f_0 = \frac{1}{1,1 \times 10000 \times 22 \times 10^{-9}}$$

$$f_0 = 4,13kHz$$

Alarm Oscillator (NE555) Role:

The NE555 is configured as an astable oscillator that drives the buzzer but only activates when the infrared beam is lost. Under normal conditions, the PLL output keeps the 555 disabled. If the beam is interrupted, the PLL changes its output state, triggering the NE555. The 555 then begins oscillating and generates a square-wave signal to power the buzzer.

- Beam present: NE555 disabled → no sound
- Beam interrupted: NE555 enabled → buzzer sounds

Key formula for 555 Astable :

$$f = \frac{1}{0,693(R_A + 2R_B)C_1}$$



From schematic:

$$R_{A(8)} = 1k\Omega$$

$$R_{B(7)} = 100k\Omega$$

$$C_1 = 47 \times 10^{-9}F$$

Calculate the Audible frequency of IC2 (NE 555 Astable):

$$f = \frac{1}{0,693(R_A + 2R_B)C_1}$$
$$f = \frac{1}{0,693 \times ((1000) + (2 \times 100000)) \times 47 \times 10^{-9}}$$
$$f = 152,74Hz$$

The NE555 oscillator is configured to generate a frequency of approximately 400 Hz. Based on the standard astable calculation, the recommended timing components are $R_{A(8)} = 17k\Omega$ and $R_{B(7)} = 30k\Omega$, along with the selected capacitor. These values ensure a stable square-wave output suitable for driving the buzzer.

The exact frequency can be verified using the NE555 astable formula:

$$f_n = \frac{1}{0,693(R_A + 2R_B)C_1}$$
$$f_n = \frac{1}{0,693 \times (17000 + 2 \times 30000) \times 47 \times 10^{-9}}$$
$$f_n \cong 398,72Hz$$

This configuration guarantees the oscillator operates at the desired frequency when triggered by the PLL output.